

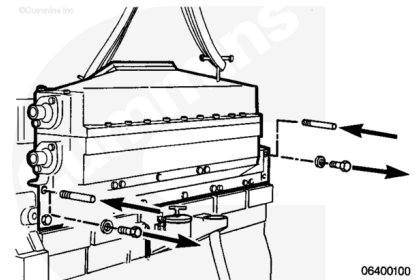
Trainings ▾

Install

QSK45 Engines

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Use guide studs long enough to protrude beyond the installed part.

Install two guide studs in the lower row of capscrew holes in the cylinder heads.

Install the new intake gaskets, with the raised bead of the gasket toward the cylinder head, to the manifold with screws.

Use Lubriplate™ 105 to lubricate the aftercooler element spouts.

Lightly tap the aftercooler assembly into location so the core spouts are positioned in the water connector.

Install the aftercooler and the top row of capscrews.

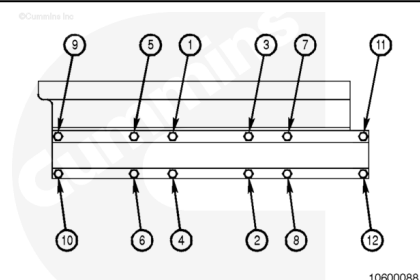
Tighten the capscrews **only** enough to hold the assembly on the engine.

Remove the guide studs. Install the bottom row of capscrews.

Tighten the intake manifold mounting capscrews in the sequence shown.

Torque Value:

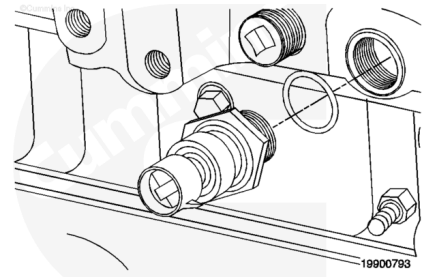
QSK45 Intake Manifold Capscrews 45 n•m [33 ft-lb]



Install the intake manifold pressure and temperature sensors.

Torque Value:

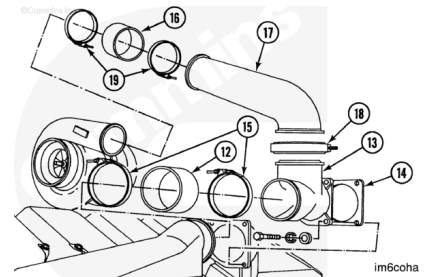
Intake Manifold Pressure and Temperature Sensors 15 n•m [133 in-lb]



Position the air crossover and the hose (16) on the turbocharger. Install the V-band clamp (18) on the air crossover and the T-connection.

Install the hose clamps (15) between the T-connection (13) and the aftercooler.

Install the T-connection and gasket (14) with four capscrews. Install the hose on the connection.

**Torque Value:**

Aftercooler Connection T-Mounting Capscrews 45 n•m [33 ft-lb]

Install the hose clamps (19 and 15) between the air crossover (17) and the turbocharger.

Torque Value:

Intake Air Systems Hose Clamps 9 n•m [80 in-lb]

Lubricate the seal with clean engine oil.

Install the seal on the retainer.

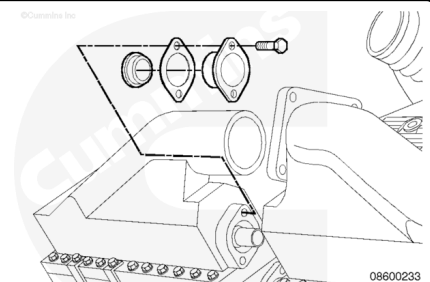
Lubricate the outer diameter of the water bosses on the aftercooler element with Lubriplate™ 105.

Slide the gasket on the water bosses.

Slide the seal and the retainer on the tube.

Install and tighten the capscrews.

Torque Value: 45 n•m [33 ft-lb]



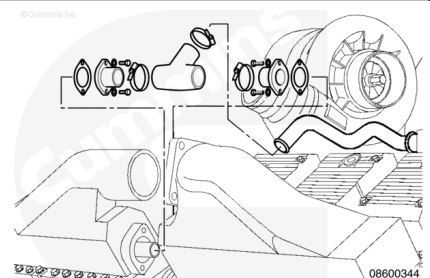
Note : Procedures for one side of the engine are shown. Procedures for the other side are similar.

Install the two rubber T-connections on the aftercooler tubes.

Install the three hose clamps on the rubber T-connections.

Tighten the hose clamps.

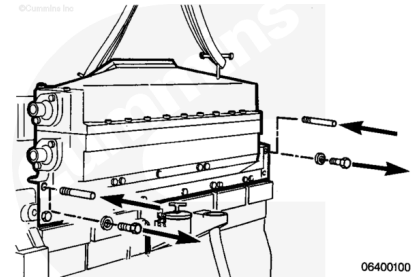
Torque Value: 5 n•m [44 in-lb]



QSK60 Engines

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Use guide studs long enough to protrude beyond the installed part.

Install two guide studs in the lower row of capscrew holes in the cylinder heads.

Install the new intake gaskets, with the raised bead of the gasket toward the cylinder head, to the aftercooler housing with screws.

Use Lubriplate™ 105 to lubricate the aftercooler element core spouts.

Lightly tap the aftercooler assembly into location so the core spouts are positioned in the water connector.

Install the aftercooler and the top row of capscrews.

Tighten the capscrews **only** enough to hold the assembly on the engine.

Remove the guide studs. Install the bottom row of capscrews.

Tighten the aftercooler assembly mounting capscrews in the sequence shown.

Torque Value:

Aluminum Aftercooler Assembly 45 n•m [33 ft-lb]

Torque Value:

Cast Iron Aftercooler Assembly 65 n•m [48 ft-lb]

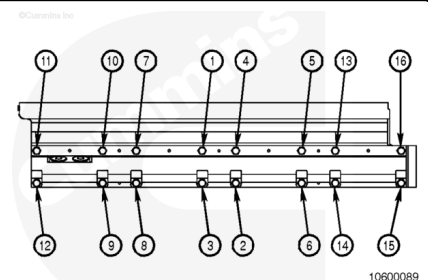
After 20 to 30 minutes, tighten the aftercooler assembly capscrews again in the sequence shown.

Torque Value:

Aluminum Aftercooler Assembly 45 n•m [33 ft-lb]

Torque Value:

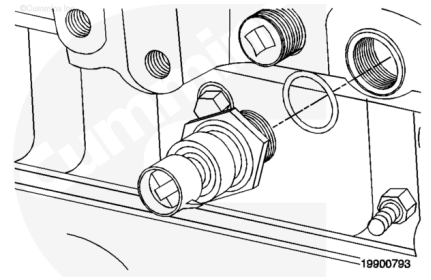
Cast Iron Aftercooler Assembly 65 n•m [48 ft-lb]



Install the intake manifold pressure and temperature sensors.

Torque Value:

Intake Manifold Pressure and Temperature Sensors 15 n•m [133 in-lb]

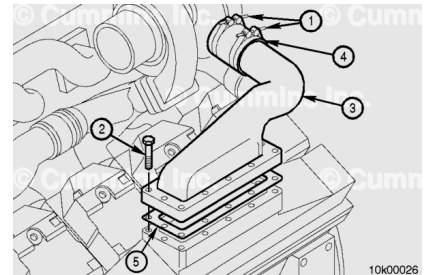


Note : The QSK60 engine contains an individual air crossover for each aftercooler. The configuration of the left-bank front and right-bank rear air crossovers are different from those used on the right-bank front and left-bank rear, but the installation procedures are the same.

Note : Engines using high pressure ratio compressor turbochargers use different air crossovers to verify alignment. Verify the same parts are installed as removed.

Position the air crossover, gasket, and hose on the turbocharger and aftercooler.

Install the two hose clamps (1) between the turbocharger and the air crossover (3). Verify the hump hose is installed square and even.

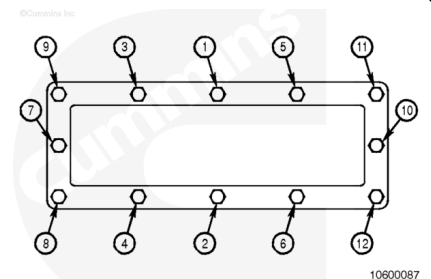
**Torque Value:**

Intake Air System Hose Clamps 9 n•m [80 in-lb]

Install and tighten the twelve capscrews on the air crossover in the pattern shown.

Torque Value:

Air Crossover Mounting Capscrews 45 n•m [33 ft-lb]



Lubricate the seal with clean engine oil.

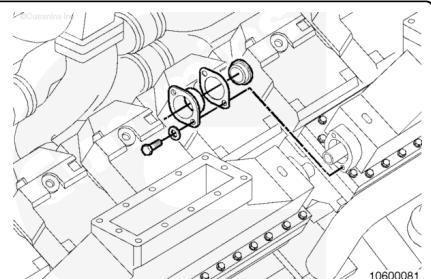
Install the seal on the retainer.

Lubricate the outer diameter of the water bosses on the aftercooler element with Lubriplate™ 105.

Slide the gasket on the water bosses.

Slide the seal and the retainer on the tube.

Install and tighten the capscrews.



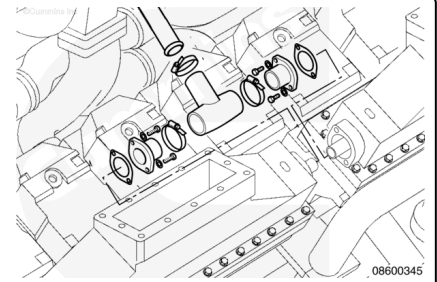
Torque Value: 45 n•m [33 ft-lb]

Install the two rubber T-connections on the four aftercooler tubes.

Install three hose clamps on the rubber T-connections.

Tighten the hose clamps.

Torque Value: 5 n•m [44 in-lb]



Last Modified: 01-Mar-2023
